

COMIC CRAFT PROJECT DOCUMENTATION

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Project Title: ComicCraft-AI

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1. Introduction:

Introduce ComicCraft as an AI-based web application that converts a user's story idea into a complete comic with story, dialogue, illustrations, and PDF export.

2. problem Statement:

Explain the difficulty of creating comics manually, such as writing stories, creating illustrations, arranging panels, and preparing the final comic.

3.Objectives:

Explain the main goals: generate stories, create dialogues, generate images, arrange comic panels, and export the final comic as a PDF.

4.Character name :

Maya

5.Role:

Protagonist / Main Character

6.Discription:

It is a mysterious time-related adventure where Maya investigates a strange clock tower, discovers a photograph from the future, and races against time to uncover its secret.

7.Scope of the projects :

Describe how ComicCraft can help non-artists, storytellers, and hobbyists create comics easily from simple prompts.

8. Technologies Used :

Explain the technologies used: Python, FastAPI, HTML, Jinja2, Gemini APIs.

9.System Requirements :

Mention the software and environment required, including Python, required libraries, API keys, virtual environment, and a web browser.

10. System Architecture:

Explain the three main parts: Frontend → FastAPI Backend → AI Services. The frontend collects inputs, FastAPI processes them, and AI models generate the story and images.

11.Working Methodology:

Explain the complete process: user enters a prompt → Gemini generates the outline → detailed story/dialogue is created → images are generated → panels are arranged → comic is exported.

12.AI Models Used:

Explain

Explain Gemini Flash for the 5-panel outline, Gemini Pro for detailed narration/dialogue, and Stable Diffusion for comic illustrations.

13.System Modules:

Describe

Describe the major modules: story generation, image generation, layout building, FastAPI backend, frontend, and PDF export.

14.Story and Comic Panel Generation:

Explain how the user's prompt is converted into a structured 5-panel outline containing panel title, scene description, and image prompt, followed by narration and dialogue.

15.AI Image Generation:

Explain how Stable Diffusion converts the generated image prompts into comic illustrations. Your “Outside the Abandoned Castle” image can be shown here as an example.

16.Comic Preview and PDF Export:

Explain how generated panels are displayed with images, descriptions, captions, and narration, and how the complete comic is converted into a downloadable PDF.

17.Testing and Results:

Explain testing of the application, API endpoints, image generation, comic generation, preview, and PDF export. The project is tested locally using the FastAPI server.

18.Conclusion and Future Enhancements:

Summarize how ComicCraft makes AI-powered comic creation easier. Future improvements can include user profiles, comic libraries, and longer multi-page stories.